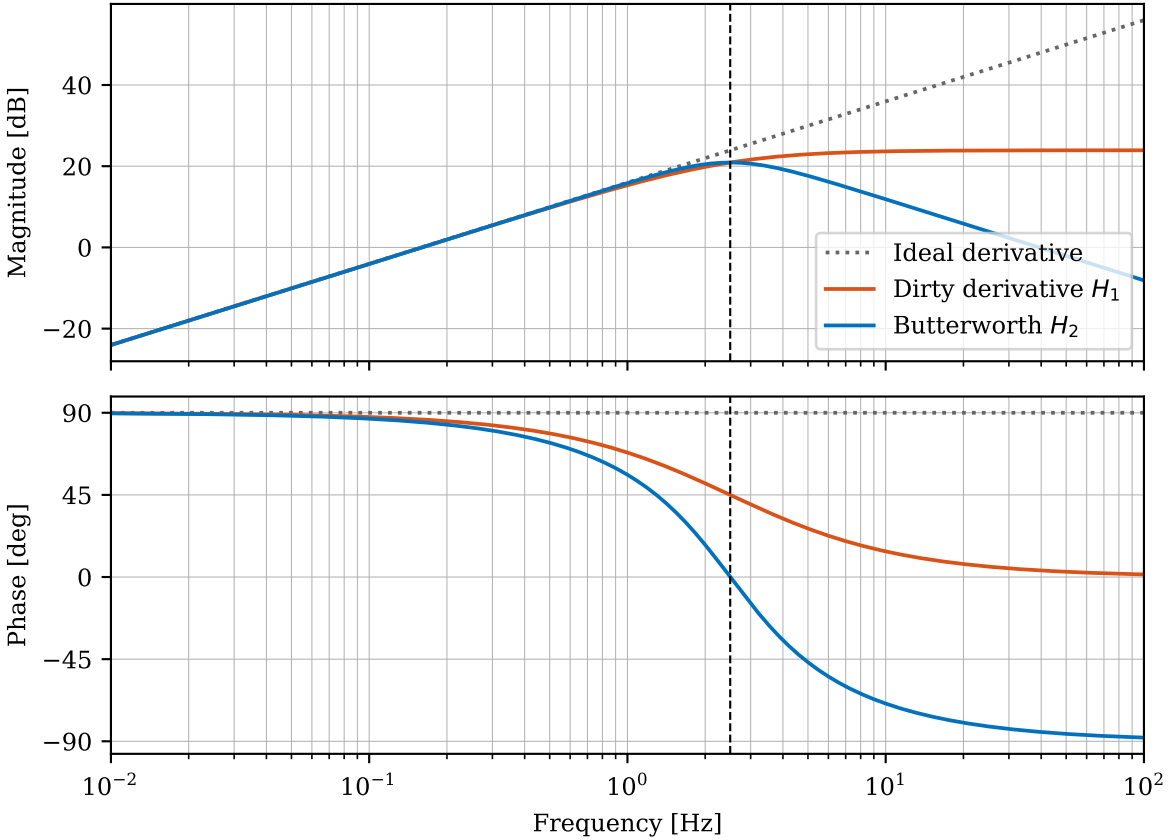
Magnitude [dB]
40
20
0
−20
Ideal derivative
Dirty derivative H_1
Butterworth H_2
Phase [deg]
90
45
0
−45
−90
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency [Hz]